

Eyal Weiss

Curriculum Vitae

Postdoctoral Researcher, Robotics Institute, Carnegie Mellon University

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Summary

Postdoctoral researcher in AI and robotics working on heuristic search, automated planning and motion planning, with a focus on planning with uncertain or expensive-to-compute action costs and on combinatorial optimization for multi-agent systems. 5 journal and 10 conference papers; SoCS 2025 Best Paper Award; Adams Fellow.

Research Interests

Artificial intelligence, robotics, heuristic search, automated planning, motion planning, multi-agent planning, combinatorial optimization; background in control theory and dynamical systems.

Research Positions

Carnegie Mellon University, Pittsburgh, PA, USA

Aug 2026 – present

Postdoctoral Researcher, Robotics Institute

- Topics: AI planning, robotics, heuristic search, motion planning, combinatorial optimization
- Hosts: [Prof. Maxim Likhachev \(Search-based Planning Laboratory\)](#) and [Prof. Jiaoyang Li \(ARCS Lab\)](#)

Technion – Israel Institute of Technology, Haifa, Israel

Oct 2024 – Aug 2026

Postdoctoral Scholar, Faculty of Computer Science

- Topics: AI planning, robotics, heuristic search, motion planning
- Host: [Prof. Oren Salzman \(Computational Robotics Lab\)](#)

Tel Aviv University, Tel Aviv, Israel

Apr 2018 – Sep 2019

Research Assistant, School of Electrical Engineering

- Topics: control theory, dynamical systems, Boolean networks
- Supervisor: [Prof. Michael Margaliot](#)

Education

Bar-Ilan University, Ramat Gan, Israel

Oct 2019 – Sep 2024

Ph.D., Computer Science

- Thesis: *Graph Search and Automated Planning with Multiple Cost Estimators*
- Advisor: [Prof. Gal A. Kaminka](#)

Tel Aviv University, Tel Aviv, Israel

Mar 2018

M.Sc., Electrical Engineering, Cum Laude

- Thesis: *Minimal Controllability and Minimal Observability of Conjunctive Boolean Networks*
- Advisor: [Prof. Michael Margaliot](#)

B.Sc., Electrical Engineering, Tel Aviv University, Summa Cum Laude

Oct 2016

Honours and Awards

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| • Best Paper Award , International Symposium on Combinatorial Search (SoCS) | 2025 |
| • Adams Fellowship (Israel Academy of Sciences and Humanities) for outstanding doctoral students | 2022 – 2024 |
| • Excellence Scholarship for research students, Computer Science Department, Bar-Ilan University | 2023 |
| • Milgat Hanasi (President's Scholarship) for excellent Ph.D. students | 2021 – 2023 |
| • Shlomo Shmeltzer Institute for Smart Transportation Scholarship | 2017 |
| • Excellent Student Prize for M.Sc. students, Faculty of Engineering, Tel Aviv University | 2017 |
| • Velger Prize for excellence in graduate studies in Systems and Control | 2017 |
| • Final Prize for excellent students in Electrical Engineering | 2015 |
| • Freescale Scholarship for excellent students in Electrical Engineering | 2014 |
| • Dean's Honors, every year of the B.Sc. | 2012 – 2016 |

Journal Publications

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1. H. Peer, E. Weiss, R. Alterovitz and O. Salzman, “Generalizing Multi-Objective Search via Objective-Aggregation Functions”, *IEEE Robotics and Automation Letters (RA-L)*, accepted, 2026.
 2. E. Weiss and M. Margaliot, “A Generalization of Linear Positive Systems with Applications to Nonlinear Systems: Invariant Sets and the Poincaré–Bendixson Property”, *Automatica*, 123, 109358, 2021.
 3. E. Weiss and M. Margaliot, “Is My System of ODEs k-Cooperative?”, *IEEE Control Systems Letters*, 5(1):73–78, 2021.
 4. E. Weiss and M. Margaliot, “A Polynomial-Time Algorithm for Solving the Minimal Observability Problem in Conjunctive Boolean Networks”, *IEEE Transactions on Automatic Control*, 64(7):2727–2736, 2019.
 5. E. Weiss and M. Margaliot, “Output Selection and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm”, *IEEE Control Systems Letters*, 3(1):210–215, 2019.
 6. E. Weiss, M. Margaliot and G. Even, “Minimal Controllability of Conjunctive Boolean Networks is NP-Complete”, *Automatica*, 92:56–62, 2018.

Conference Publications

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1. Y. Wang, E. Weiss, B. Mu and O. Salzman, “Bidirectional Search while Ensuring Meet-In-The-Middle via Effective and Efficient-to-Compute Termination Conditions”, *Proc. International Joint Conference on Artificial Intelligence (IJCAI)*, 8987–8995, 2025.
 2. Y. Sherma, E. Weiss and O. Salzman, “From Agent Centric to Obstacle Centric Planning: A Makespan-Optimal Algorithm for the Multi-Agent Warehouse Rearrangement Problem”, *Proc. International Symposium on Combinatorial Search (SoCS)*, 136–144, 2025. **Best Paper Award.**
 3. D. Gnad, L. Alon, E. Weiss and A. Shleyfman, “PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning”, *Proc. AAAI Conference on Artificial Intelligence*, 26507–26515, 2025.
 4. E. Weiss, A. Felner and G. A. Kaminka, “Tightest Admissible Shortest Path”, *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, 643–652, 2024.
 5. E. Weiss, A. Felner and G. A. Kaminka, “A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates”, *Proc. European Conference on Artificial Intelligence (ECAI)*, 2607–2614, 2023.
 6. E. Weiss and G. A. Kaminka, “Planning with Multiple Action-Cost Estimates”, *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, 427–437, 2023.
 7. E. Weiss and M. Margaliot, “A Generalization of Linear Positive Systems”, *27th Mediterranean Conference on Control and Automation (MED)*, 340–345, 2019.
 8. E. Weiss and M. Margaliot, “A Generalization of Smillie's Theorem on Strongly Cooperative Tridiagonal Systems”, *IEEE Conference on Decision and Control (CDC)*, 3080–3085, 2018.

9. **E. Weiss** and M. Margaliot, “Observability Analysis and Observer Design for Boolean Control Networks: A Sub-Optimal Polynomial-Complexity Algorithm”, *IEEE International Conference on the Science of Electrical Engineering (ICSEE)*, 2018.

Workshop Publications

1. D. Gnad, L. Alon, **E. Weiss** and A. Shleyfman, “PDBs Go Numeric: Pattern-Database Heuristics for Simple Numeric Planning”, *ICAPS’24 Workshop on Heuristics and Search for Domain-Independent Planning (HSDIP)*, 2024.
2. **E. Weiss**, A. Felner and G. A. Kaminka, “A Generalization of the Shortest Path Problem to Graphs with Multiple Edge-Cost Estimates”, *ICAPS’23 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2023.
3. **E. Weiss** and G. A. Kaminka, “Planning with Dynamically Estimated Action Costs”, *ICAPS’22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.
4. **E. Weiss** and G. A. Kaminka, “Position Paper: Online Modeling for Offline Planning”, *ICAPS’22 Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS)*, 2022.

Invited Talks

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| • Search-based Planning Laboratory (SBPL), Carnegie Mellon University | May 2025 |
| • Computational Robotics Lab (CRL), Technion | Jun 2024 |
| • Adams Fellowship Conference, Israel Academy of Sciences and Humanities, Jerusalem | Feb 2023 |
| • Bar-Ilan Symposium on Foundations of AI (BISFAI), Ramat Gan | Feb 2023 |
| • Israeli Association for Artificial Intelligence (IAAI) Symposium, Haifa | Jun 2022 |

Software

- [PlanDEM](#) – domain-independent planner for dynamically estimated action models (author)
- [MEET](#) – bidirectional search (IJCAI 2025); being integrated into [OMPL](#)
- [Warehouse Rearrangement](#) – reference implementation (SoCS 2025)
- [Numeric Fast Downward](#) – numeric pattern-database heuristics (AAAI 2025)
- [CSNA](#) – cost-sensitive neighborhood aggregation for graph neural networks

Service

Reviewer, conferences: ICAPS, AAAI, IJCAI, RSS, ICANN

Reviewer, journals: IEEE Robotics and Automation Letters; Automatica; IEEE Transactions on Automatic Control; European Journal of Control; IEEE Transactions on Network Science and Engineering; Optimization and Engineering

Organizing committee

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| • ICAPS Workshop on Reliability In Planning and Learning (RIPL) | 2026 |
| • ICAPS Workshop on Heuristics and Search for Domain-independent Planning (HSDIP) | 2025 |
| • ICAPS Workshop on Reliable Data-Driven Planning and Scheduling (RDDPS) | 2023, 2024 |

Session chair

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| • European Conference on Artificial Intelligence (ECAI) | 2023 |
| • IEEE Conference on Decision and Control (CDC) | 2018 |

Student Supervision and Mentoring

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| • Research mentoring of graduate students at CRL, Technion, leading to joint publications (IJCAI 2025, SoCS 2025) | 2024 – 2026 |
| • Supervision of Computer Science graduation projects in heuristic search, Technion | 2025 – 2026 |
| • Supervision of gifted high-school students' research projects, Alpha program, Bar-Ilan University | 2022 – 2024 |
| • Supervision of Electrical Engineering graduation projects in autonomous driving with mini-scale cars, Tel Aviv University | 2018 – 2019 |

Teaching Experience

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| Bar-Ilan University , Department of Computer Science – <i>Teaching Assistant</i> | 2020 – 2024 |
| • Artificial Intelligence (2024); Statistical Methods in Computer Science (2023); Introduction to Intelligent, Smart and Cognitive Systems (2021–2022); Object Oriented Programming (2021); Computer Structure (2020–2022) | |
| Tel Aviv University , School of Electrical Engineering – <i>Teaching Assistant and Laboratory Instructor</i> | 2016 – 2020 |
| • Stochastic Processes (2019–2020); Signals and Systems (2017–2020); Random Signals and Noise (2017–2020); Optimal Control (2016–2017) | |
| • Laboratory of Advanced Control (2016–2020) | |

Professional Experience

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| Mellanox Technologies (now part of NVIDIA), Israel – <i>Network Architecture Intern</i> | 2015 |
| Mellanox Technologies – <i>Chip Design Intern</i> | 2013 – 2014 |

Skills and Languages

- Programming: C, C++, Python, Java, MATLAB, Bash; comfortable picking up additional languages as needed
- Laboratory: robotic systems, electronics and low-level controllers
- Languages: Hebrew (native), English (full professional proficiency)

Community and Volunteering

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| • Tutor, economic empowerment program of “Misdar Dorshei Tov” | 2018 – 2022 |
| • Technical editor, “Good to Know” project, Tel Aviv University | 2017 |
| • Personal mentor for youth at risk, “Shimony” Children’s Home | 2012 – 2014 |